Basic Networking Skills for Digital Forensics

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File Acquisition and Integrity

ield	Value
Original File name	memdumpWin7.mem
Source	*wget https://www.dropbox.com/s/x41f2lyhlrixts6/memdumpWin7.mem*
File size	1073676288 bytes
Download date	2026-10-18 20:35:39
SHA256 (Original)	500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e
Tools Version	Volatility Foundation Volatility Framework 2.6.1
Working copy	Workin/memdumpWin7.mem
SHA256 (working copy)	500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e
note	Hashes matches the original file succesfully

1. Preserving Evidence (Question 1)

1.1 Case Folder Creation

```
┌──(root㉿kali)-[~]
└─# mkdir -p ~/SBT-DF204-Case3/{evidence,working,exported,reports,screenshots}
```

Figure 1 (EO1). mkdir -p ~/SBT-DF204-Case3/{evidence,working,exported, reports and screenshots} – creates the subfolders for evidence and file storage throughout the studies.

1.2 Downloading the Evidence File

```
┌──(kali㉿kali)-[~/SBT-DF204-Case3]
└─$ wget https://www.dropbox.com/s/x41f2lyhlrixts6/memdumpWin7.mem

--2026-10-18 20:07:17--  https://www.dropbox.com/s/x41f2lyhlrixts6/memdumpWin7.mem
Resolving www.dropbox.com (www.dropbox.com)... 162.125.68.18, 2620:100:6024:18::a27d:4412
Connecting to www.dropbox.com (www.dropbox.com)|162.125.68.18|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://www.dropbox.com/scl/fi/1y68wpuiq1fsvrapq9egg/memdumpWin7.mem?rlkey=pk3thrk3p2fk9m90mywksx3s1 [following]
--2026-10-18 20:07:20--  https://www.dropbox.com/scl/fi/1y68wpuiq1fsvrapq9egg/memdumpWin7.mem?rlkey=pk3thrk3p2fk9m90mywksx3s1
Reusing existing connection to www.dropbox.com:443.
HTTP request sent, awaiting response... 302 Found
```

```
┌──(kali㉿kali)-[~/SBT-DF204-Case3]
└─$ ls -l
total 1048540
drwxrwxr-x 2 kali kali       4096 Oct 18 05:55 evidence
drwxrwxr-x 2 kali kali       4096 Oct 18 05:55 exported
-rw-rw-r-- 1 kali kali 1073676288 Oct 18 20:21 memdumpWin7.mem
drwxrwxr-x 2 kali kali       4096 Oct 18 05:55 reports
drwxrwxr-x 2 kali kali       4096 Oct 18 05:55 screenshots
drwxrwxr-x 2 kali kali       4096 Oct 18 05:55 scripts
drwxrwxr-x 2 kali kali       4096 Oct 18 05:55 working
```

Figure 2 (E02). wget https://www.dropbox.com/s/x41f2lyhlrixts6/memdumpWin7.mem followed by ls -l command to verify downloaded file.

```
┌──(kali㉿kali)-[~/SBT-DF204-Case3]
└─$ cp --preserve=timestamps evidence/memdumpWin7.mem  working/memdumpWin7.mem
```

Figure 3 (E03) cp --preserve=timestamps evidence/memdumpWin7.mem working/memdumpWin7.mem, original file preserved to avoid tempering.

1.3 Verifying Integrity

```
┌──(kali㉿kali)-[~/SBT-DF204-Case3]
└─$ sha256sum evidence/memdumpWin7.mem | tee reports/memdumpWin7.mem_original.txt
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e  evidence/memdumpWin7.mem

┌──(kali㉿kali)-[~/SBT-DF204-Case3]
└─$

┌──(kali㉿kali)-[~/SBT-DF204-Case3]
└─$ sha256sum working/memdumpWin7.mem | tee reports/memdumpWin7.mem_working.txt
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e  working/memdumpWin7.mem

┌──(kali㉿kali)-[~/SBT-DF204-Case3]
└─$
```

Figure 4 (E04) sha256sum evidence/memdumpWin7.mem | tee reports/memdumpWin7.mem_original.txt – records the SHA256 hash of the original evidence file before starting any analysis.

2. Identifying and validating the profile (Question 2)

```
          Suggested Profile(s) : Win7SP1×86_23418, Win7SP0×86, Win7SP1×86_24000, Win7SP1×86
                     AS Layer1 : IA32PagedMemoryPae (Kernel AS)
                     AS Layer2 : FileAddressSpace (/home/kali/SBT-DF204-Case3/working/memdumpWin7.mem)
                      PAE type : PAE
                           DTB : 0×185000L
                          KDBG : 0×82972c68L
        Number of Processors : 1
     Image Type (Service Pack) : 1
                KPCR for CPU 0 : 0×82973d00L
             KUSER_SHARED_DATA : 0×ffdf0000L
           Image date and time : 2019-01-06 15:09:06 UTC+0000
     Image local date and time : 2019-01-06 07:09:06 -0800

┌──(venv)─(kali㉿kali)-[~/SBT-DF204-Case3/volatility]
└─$
```

*Figure 5 (E05) profile identified and successfully validated using (python2 vol.py -f ~/SBT-DF204-Case3/working/memdumpWin7.mem imageinfo | tee ~/SBT-DF204-Case3/reports/imageinfo.txt) and saved reports have been recorded with Win7SP1*86_23418*

suggested profile, 1 number of processors, 2019-01-06 15:09:06 UTC+0000 and local time 2019-01-06 07:09:06 -0800

2.1 profile (Win7SP1x86_23418) Pslist

```
0×859bab00 taskeng.exe          1776    880      6       85      0      0 2019-01-06 15:02:48 UTC+0000
0×8500bd00 cygrunsrv.exe        1984   1680      0 ------          0      0 2019-01-06 15:02:49 UTC+0000   2019-01-06 15:02:49 UTC+0000
0×84202030 conhost.exe          2000    300      2       33      0      0 2019-01-06 15:02:49 UTC+0000
0×84203998 sshd.exe             2016   1984      4      100      0      0 2019-01-06 15:02:49 UTC+0000
0×8501cce8 GoogleUpdate.e        192   1776      4      109      0      0 2019-01-06 15:02:49 UTC+0000
0×85b11398 sppsvc.exe           1036    432      4      143      0      0 2019-01-06 15:02:50 UTC+0000
0×85a07030 svchost.exe          2072    432      5       91      0      0 2019-01-06 15:02:50 UTC+0000
0×857ce6d0 rundll32.exe         2320    552      4       85      1      0 2019-01-06 15:02:52 UTC+0000
0×84fddd28 dwm.exe              2512    828      3       70      1      0 2019-01-06 15:02:54 UTC+0000
0×85bba860 explorer.exe         2524   2500     32      952      1      0 2019-01-06 15:02:54 UTC+0000
0×85c7f030 VBoxTray.exe         2632   2524     13      140      1      0 2019-01-06 15:02:55 UTC+0000
0×85cb9820 SearchIndexer.       2960    432     13      607      0      0 2019-01-06 15:03:01 UTC+0000
0×85e287f0 WUDFHost.exe         3300    828      8      205      0      0 2019-01-06 15:03:07 UTC+0000
0×84fa2ad0 msiexec.exe          3748    432      4      174      0      0 2019-01-06 15:03:34 UTC+0000
0×841eb658 WmiPrvSE.exe         4032    552      5      114      0      0 2019-01-06 15:03:54 UTC+0000
0×8583a030 chrome.exe           2388   2524     33      819      1      0 2019-01-06 15:04:30 UTC+0000
0×851e8520 chrome.exe           1280   2388      7       82      1      0 2019-01-06 15:04:30 UTC+0000
0×850316b8 chrome.exe           2380   2388      2       57      1      0 2019-01-06 15:04:30 UTC+0000
0×852845c8 svchost.exe          2416    432     10      139      0      0 2019-01-06 15:04:30 UTC+0000
0×842b4d28 chrome.exe           2912   2388      9      159      1      0 2019-01-06 15:04:32 UTC+0000
0×858be710 svchost.exe          3784    432     14      370      0      0 2019-01-06 15:04:51 UTC+0000
0×8431fb08 cmd.exe              3996   2524      1       22      1      0 2019-01-06 15:06:11 UTC+0000
0×842ddd28 conhost.exe          3988    348      2       52      1      0 2019-01-06 15:06:11 UTC+0000
0×842eb6b0 chrome.exe            976   2388     14      198      1      0 2019-01-06 15:06:23 UTC+0000
```

Figure 6; (E06) The first profile (Win7SP1x86_23418) showing the Offset(V) Name, PID, PPID, UTC+ Started time and Exit

2.2 Profile (Win7SP1x86_23418) hivelist/registry keys

```
**************************************************
Registry: \REGISTRY\MACHINE\BCD00000000
Key name: NewStoreRoot (S)
Last updated: 2019-01-06 18:02:44 UTC+0000

Subkeys:
  (S) Description
  (S) Objects

Values:
**************************************************
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: CMI-CreateHive{6A1C4018-979D-4291-A7DC-7AED1C75B67C} (S)
Last updated: 2019-01-06 15:02:48 UTC+0000

Subkeys:
  (S) AppEvents
  (S) Console
  (S) Control Panel
  (S) Environment
  (S) EUDC
  (S) Keyboard Layout
  (S) Network
  (S) Printers
  (S) Software
  (S) System
  (V) Volatile Environment

Values:
**************************************************
Registry: \??\C:\System Volume Information\Syscache.hve
Key name: {a5b8a9ac-608c-11e5-a266-8af578093dd2} (S)
Last updated: 2019-01-06 15:02:50 UTC+0000

Subkeys:
  (S) DefaultObjectStore
```

Figure 7 (E07) the profile (Win7SP1x86_23418) hive and registry keys \??\c:\Users\IEUser\ntuser.dat with 2019-01-06 15:02:48 UTC+0000 time.

```
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: TimeZoneInformation (S)
Last updated: 2019-01-06 16:31:39 UTC+0000

Subkeys:

Values:
REG_DWORD     Bias            : (S) 480
REG_DWORD     DaylightBias    : (S) 4294967236
REG_SZ        DaylightName    : (S) @tzres.dll,-211
REG_BINARY    DaylightStart   : (S)
0x00000000  00 00 03 00 02 00 02 00 00 00 00 00 00 00 00 00   ................
REG_DWORD     StandardBias    : (S) 0
REG_SZ        StandardName    : (S) @tzres.dll,-212
REG_BINARY    StandardStart   : (S)
0x00000000  00 00 0b 00 01 00 02 00 00 00 00 00 00 00 00 00   ................
REG_SZ        TimeZoneKeyName : (S) Pacific Standard Time
REG_DWORD     DynamicDaylightTimeDisabled : (S) 0
REG_DWORD     ActiveTimeBias  : (S) 480

┌──(venv)─(kali㉿kali)-[~/SBT-DF204-Case3/volatility]
└─$
```

python2 vol.py -f ~/SBT-DF204-Case3/working/memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K "ControlSet001\Control\TimeZoneInformation" | tee ~/SBT-DF204-Case3/reports/timezone.txt

3. User and account artefacts (Question 3)

```
Virtual    Physical   Name
---------- ---------- ----
0x87c10370 0x27fd5370 [no name]
0x87c1c008 0x27fa3008 \REGISTRY\MACHINE\SYSTEM
0x87c459c8 0x27d8e9c8 \REGISTRY\MACHINE\HARDWARE
0x889c0430 0x1285a430 \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT
0x8b46d008 0x16d8c008 \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT
0x8b5d39b0 0x120049b0 \??\C:\Users\IEUser\ntuser.dat
0x97104008 0x1c45f008 \SystemRoot\System32\Config\SECURITY
0x9711d9c8 0x1e6a09c8 \SystemRoot\System32\Config\SOFTWARE
0x981f9008 0x1b446008 \SystemRoot\System32\Config\DEFAULT
0x98209518 0x1f3a7518 \SystemRoot\System32\Config\SAM
0x982d55c0 0x1e17c5c0 \REGISTRY\MACHINE\BCD00000000
0xa620b008 0x1997b008 \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa65269c8 0x17ad09c8 \??\C:\Users\sshd_server\ntuser.dat
0xa6539260 0x0f690260 \??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa7d2a9c8 0x01c469c8 \??\C:\System Volume Information\Syscache.hve
```

Figure 8 (E08) python2 vol.py -f ~/SBT-DF204-Case3/working/memdumpWin7.mem --profile=Win7SP1x86_23418 hivelist. the virtual offset of each user's NTUSER.DAT and UsrClass.dat have been recored for MountPoints2 and ShellBags.

3.1 who is associating with the device

```
Legend: (S) = Stable   (V) = Volatile

_____________________________

Registry: \SystemRoot\System32\Config\SOFTWARE
Key name: ProfileList (S)
Last updated: 2015-09-21 09:50:52 UTC+0000

Subkeys:
  (S) S-1-5-18
  (S) S-1-5-19
  (S) S-1-5-20
  (S) S-1-5-21-1716914095-909560446-1177810406-1000
  (S) S-1-5-21-1716914095-909560446-1177810406-1002

Values:
REG_EXPAND_SZ ProfilesDirectory : (S) %SystemDrive%\Users
REG_EXPAND_SZ Default           : (S) %SystemDrive%\Users\Default
REG_EXPAND_SZ Public            : (S) %SystemDrive%\Users\Public
REG_EXPAND_SZ ProgramData       : (S) %SystemDrive%\ProgramData
```

Figure 9 (E09) python2 vol.py -f ~/SBT-DF204-Case3/working/memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K "Microsoft\Windows NT\CurrentVersion\ProfileList". The associate of the device Registry: \systemRoot\System32\Config\SOFTWARE with multiple subkeys.

SSID Format

SID	Name	Description
S-1-5-18	Local System	A service account that is used by the operating system.
S-1-5-19	NT Authority	Local Service
S-1-5-20	NT Authority	Network Service
S-1-5-21	Admin/Guest	A user account for the system administrator
S-1-5-22	Admin/Guest	A user account for the system administrator

4. Processes and network evidence (Question 4)

python2 vol.py -f ~/SBT-DF204-Case3/working/memdumpWin7.mem --profile=Win7SP1x86_23418 pslist

```
0x85ae3030 vmicsvc.exe           1388    432      5      105      0      0 2019-01-06 15:02:48 UTC+0000
0x85ae9030 taskhost.exe          1412    432      8      145      1      0 2019-01-06 15:02:48 UTC+0000
0x85af6d28 vmicsvc.exe           1440    432      3       66      0      0 2019-01-06 15:02:48 UTC+0000
0x85b06d28 vmicsvc.exe           1492    432      4       80      0      0 2019-01-06 15:02:48 UTC+0000
0x850553f0 vmicsvc.exe           1548    432      4       81      0      0 2019-01-06 15:02:48 UTC+0000
0x85b2dd28 svchost.exe           1588    432     10      147      0      0 2019-01-06 15:02:48 UTC+0000
0x859dc030 cygrunsrv.exe         1680    432      6      100      0      0 2019-01-06 15:02:48 UTC+0000
0x859a0d28 wlms.exe              1752    432      4       45      0      0 2019-01-06 15:02:48 UTC+0000
0x859bab00 taskeng.exe           1776    880      6       85      0      0 2019-01-06 15:02:48 UTC+0000
0x8500bd00 cygrunsrv.exe         1984   1680      0 --------      0      0 2019-01-06 15:02:49 UTC+0000   2019-01-06 15:02:49 UTC+0000
0x84202030 conhost.exe           2000    300      2       33      0      0 2019-01-06 15:02:49 UTC+0000
0x84203998 sshd.exe              2016   1984      4      100      0      0 2019-01-06 15:02:49 UTC+0000
0x8501cce8 GoogleUpdate.e         192   1776      4      109      0      0 2019-01-06 15:02:49 UTC+0000
0x85b11398 sppsvc.exe            1036    432      4      143      0      0 2019-01-06 15:02:50 UTC+0000
0x85a07030 svchost.exe           2072    432      5       91      0      0 2019-01-06 15:02:50 UTC+0000
0x857ce6d0 rundll32.exe          2320    552      4       85      1      0 2019-01-06 15:02:52 UTC+0000
0x84fddd28 dwm.exe               2512    828      3       70      1      0 2019-01-06 15:02:54 UTC+0000
0x85bba860 explorer.exe          2524   2500     32      952      1      0 2019-01-06 15:02:54 UTC+0000
0x85c7f030 VBoxTray.exe          2632   2524     13      140      1      0 2019-01-06 15:02:55 UTC+0000
0x85cb9820 SearchIndexer.        2960    432     13      607      0      0 2019-01-06 15:03:01 UTC+0000
0x85e287f0 WUDFHost.exe          3300    828      8      205      0      0 2019-01-06 15:03:07 UTC+0000
0x84fa2ad0 msiexec.exe           3748    432      4      174      0      0 2019-01-06 15:03:34 UTC+0000
0x841eb658 WmiPrvSE.exe          4032    552      5      114      0      0 2019-01-06 15:03:54 UTC+0000
0x8583a030 chrome.exe            2388   2524     33      819      1      0 2019-01-06 15:04:30 UTC+0000
0x851e8520 chrome.exe            1280   2388      7       82      1      0 2019-01-06 15:04:30 UTC+0000
0x850316b8 chrome.exe            2380   2388      2       57      1      0 2019-01-06 15:04:30 UTC+0000
0x852845c8 svchost.exe           2416    432     10      139      0      0 2019-01-06 15:04:30 UTC+0000
0x842b4d28 chrome.exe            2912   2388      9      159      1      0 2019-01-06 15:04:32 UTC+0000
0x858be710 svchost.exe           3784    432     14      370      0      0 2019-01-06 15:04:51 UTC+0000
0x8431fb08 cmd.exe               3996   2524      1       22      1      0 2019-01-06 15:06:11 UTC+0000
0x842ddd28 conhost.exe           3988    348      2       52      1      0 2019-01-06 15:06:11 UTC+0000
0x842eb6b0 chrome.exe             976   2388     14      198      1      0 2019-01-06 15:06:23 UTC+0000
0x84312030 FTK Imager.exe        2596   2524     14      358      1      0 2019-01-06 15:07:15 UTC+0000
0x850d5a38 dllhost.exe           2228    552      6 170...38      1      0 2019-01-06 15:09:07 UTC+0000
```

Figure 10 (E10) Win7SP1x86_23418 pslist with PID,PPIDs and UTC time.

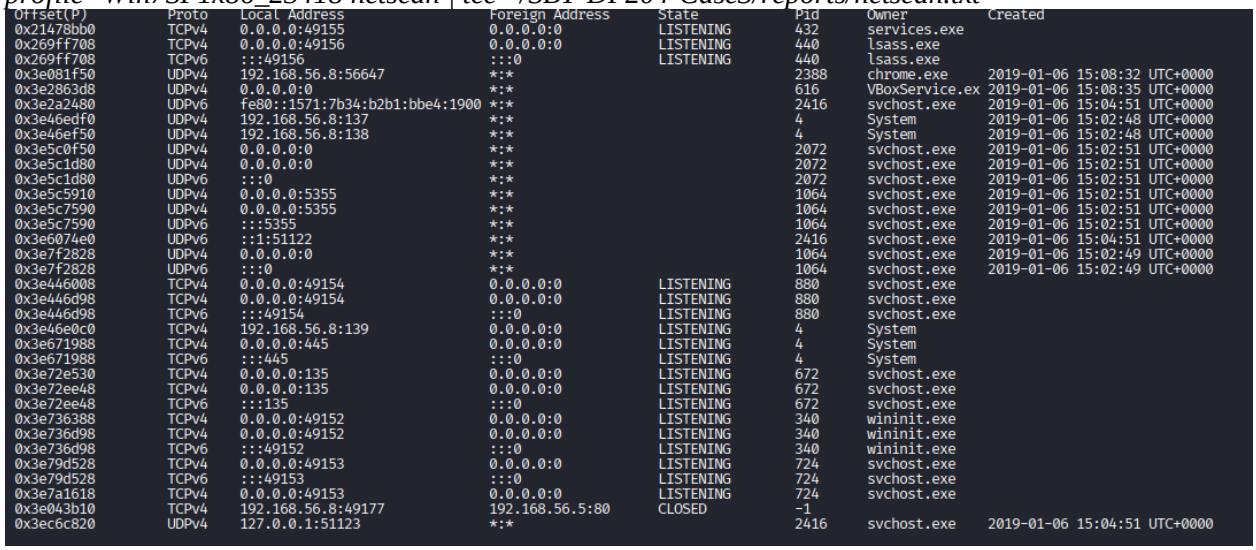
Processes	PID	PPID	Start time (UTC+0000)	Offset (V)
Winlogon.exe	388	328	2019-01-06 18:02:45	0x858a4d28

Explorer.exe	2524	2500	2019-01-06 15:02:54	0x85bba860
Chrome.exe	2388	2524	2019-01-06 15:04:30	0x8583a030
Cmd.exe	3996	2524	2019-01-06 15:06:11	0x8431fb08
FTK Imager.exe	2596	2524	2019-01-06 15:07:15	0x84312030

The identified five processes and their PIDs, PPIDs, Start times and memory offsets. Explorer.exe appeasr to be the parent of chrome.exe, cmd.exe and FTK Imager.exe.

4.2 Network Scan

python2 vol.py -f ~/SBT-DF204-Case3/working/memdumpWin7.mem --profile=Win7SP1x86_23418 netscan | tee ~/SBT-DF204-Case3/reports/netscan.txt



Offset(P)	Proto	Local Address	Foreign Address	State	Pid	Owner	Created
0x21478bb0	TCPv4	0.0.0.0:49155	0.0.0.0:0	LISTENING	432	services.exe	
0x269ff708	TCPv4	0.0.0.0:49156	0.0.0.0:0	LISTENING	440	lsass.exe	
0x269ff708	TCPv6	:::49156	:::0	LISTENING	440	lsass.exe	
0x3e081f50	UDPv4	192.168.56.8:56647	*:*		2388	chrome.exe	2019-01-06 15:08:32 UTC+0000
0x3e2863d8	UDPv4	0.0.0.0:0	*:*		616	VBoxService.ex	2019-01-06 15:08:35 UTC+0000
0x3e2a2480	UDPv6	fe80::1571:7b34:b2b1:bbe4:1900	*:*		2416	svchost.exe	2019-01-06 15:04:51 UTC+0000
0x3e46edf0	UDPv4	192.168.56.8:137	*:*		4	System	2019-01-06 15:02:48 UTC+0000
0x3e46ef50	UDPv4	192.168.56.8:138	*:*		4	System	2019-01-06 15:02:48 UTC+0000
0x3e5c0f50	UDPv4	0.0.0.0:0	*:*		2072	svchost.exe	2019-01-06 15:02:51 UTC+0000
0x3e5c1d80	UDPv4	0.0.0.0:0	*:*		2072	svchost.exe	2019-01-06 15:02:51 UTC+0000
0x3e5c1d80	UDPv6	:::0	*:*		2072	svchost.exe	2019-01-06 15:02:51 UTC+0000
0x3e5c5910	UDPv4	0.0.0.0:5355	*:*		1064	svchost.exe	2019-01-06 15:02:51 UTC+0000
0x3e5c7590	UDPv4	0.0.0.0:5355	*:*		1064	svchost.exe	2019-01-06 15:02:51 UTC+0000
0x3e5c7590	UDPv6	:::5355	*:*		1064	svchost.exe	2019-01-06 15:02:51 UTC+0000
0x3e6074e0	UDPv6	::1:51122	*:*		2416	svchost.exe	2019-01-06 15:04:51 UTC+0000
0x3e7f2828	UDPv4	0.0.0.0:0	*:*		1064	svchost.exe	2019-01-06 15:02:49 UTC+0000
0x3e7f2828	UDPv6	:::0	*:*		1064	svchost.exe	2019-01-06 15:02:49 UTC+0000
0x3e446008	TCPv4	0.0.0.0:49154	0.0.0.0:0	LISTENING	880	svchost.exe	
0x3e446d98	TCPv4	0.0.0.0:49154	0.0.0.0:0	LISTENING	880	svchost.exe	
0x3e446d98	TCPv6	:::49154	:::0	LISTENING	880	svchost.exe	
0x3e46e0c0	TCPv4	192.168.56.8:139	0.0.0.0:0	LISTENING	4	System	
0x3e671988	TCPv4	0.0.0.0:445	0.0.0.0:0	LISTENING	4	System	
0x3e671988	TCPv6	:::445	:::0	LISTENING	4	System	
0x3e72e530	TCPv4	0.0.0.0:135	0.0.0.0:0	LISTENING	672	svchost.exe	
0x3e72ee48	TCPv4	0.0.0.0:135	0.0.0.0:0	LISTENING	672	svchost.exe	
0x3e72ee48	TCPv6	:::135	:::0	LISTENING	672	svchost.exe	
0x3e736388	TCPv4	0.0.0.0:49152	0.0.0.0:0	LISTENING	340	wininit.exe	
0x3e736d98	TCPv4	0.0.0.0:49152	0.0.0.0:0	LISTENING	340	wininit.exe	
0x3e736d98	TCPv6	:::49152	:::0	LISTENING	340	wininit.exe	
0x3e79d528	TCPv4	0.0.0.0:49153	0.0.0.0:0	LISTENING	724	svchost.exe	
0x3e79d528	TCPv6	:::49153	:::0	LISTENING	724	svchost.exe	
0x3e7a1618	TCPv4	0.0.0.0:49153	0.0.0.0:0	LISTENING	724	svchost.exe	
0x3e043b10	TCPv4	192.168.56.8:49177	192.168.56.5:80	CLOSED	-1		
0x3ec6c820	UDPv4	127.0.0.1:51123	*:*		2416	svchost.exe	2019-01-06 15:04:51 UTC+0000

Figure 11 (E11) Win7SP1x86_23418 netscan with local and foreign endpoints and UTC time.

Processes	Protocol- IP address	Local endpoint	Foreign endpoint	State	Establishment time
chrome.exe	UDPv4	192.168.56.8:56647	*:*	Listenning	2019-01-06 15:08:32 UTC+0000
svchost.exe	UDPv6	::1:1900	*:*	-	2019-01-06 15:04:51 UTC+0000
services.exe	CPv4	0.0.0.0:49155	0.0.0.0	Listenning	-

The combined evidence shows chrome activity and several network endpoints, including local service-related listeners.

5. Command history (Question 5)

*python2 vol.py -f ~/SBT-DF204-Case3/working/memdumpWin7.mem --profile=*Win7SP1x86_23418 *cmdscan | tee ~/SBT-DF204-Case3/reports/cmdscan.txt*

```
CommandProcess: conhost.exe Pid: 2000
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
Cmd #29 @ 0x3200c4: 3?4?2???2
Cmd #37 @ 0x3200c4: 3?4?2???2
**************************************************
CommandProcess: conhost.exe Pid: 3988
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 @ 0x10dc40: ipconfig
Cmd #1 @ 0x107f90: cd Downloads
Cmd #2 @ 0x1147a0: copy secret_file.docx F:
Cmd #22 @ 0xff818488: ?
Cmd #25 @ 0xff818488: ?
Cmd #36 @ 0xe00c4: ?????
Cmd #37 @ 0x10cff0: ?????

┌──(venv)─(kali㉿kali)-[~/SBT-DF204-Case3/volatility]
└─$
```

```
**************************************************
ConsoleProcess: conhost.exe Pid: 3988
Console: 0xe181c0 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4
OriginalTitle: %SystemRoot%\system32\cmd.exe
Title: C:\Windows\system32\cmd.exe
AttachedProcess: cmd.exe Pid: 3996 Handle: 0x5c
----
CommandHistory: 0x1105d8 Application: ipconfig.exe Flags:
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x0
----
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 at 0x10dc40: ipconfig
Cmd #1 at 0x107f90: cd Downloads
Cmd #2 at 0x1147a0: copy secret_file.docx F:
----
Screen 0xf63b8 X:80 Y:300
Dump:
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation.  All rights reserved.

C:\Users\IEUser>ipconfig

Windows IP Configuration


Ethernet adapter Local Area Connection 3:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::1571:7b34:b2b1:bbe4%18
   IPv4 Address. . . . . . . . . . . : 192.168.56.8
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
```

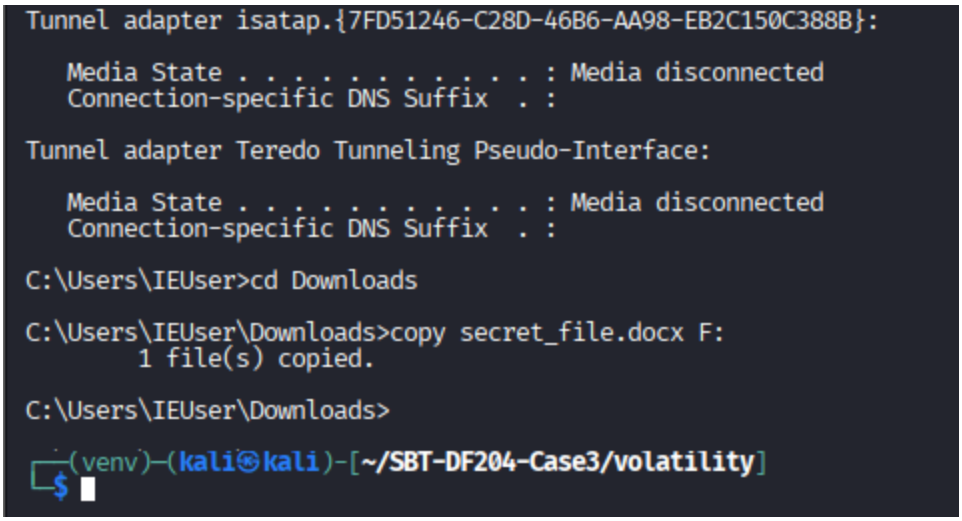


Figure 12(E12). Command History Analysis/consoles (Volatility cmdscan)

5.1 Observations

Field	Entry 1	Entry 2
CommandProcess	Conhost.exe (PID 2000)	Conhost.exe (PID 3988)
Application	Sshd.exe	cmd.exe
Cmd History offset	0x320960	0x110448
Flags	Allocated	Allocated,Reset
CommandCount/LastAdded	0/-1	3/2
CoomandCountMax	50	50
ProcessHandle	0x54	0x5c

Entry 2 (cmd.exe,PID 3988): recovered commands

i. Cmd #0: ipconfig
ii. Cmd #1: cd Downloads
iii. Cmd #2: copy secret_file.docx F:

Unreadable entries: Entry 1 (cmd #29, #37) and Entry 2 (cmd #22, #25, #37) show only ? characters

5.2 Analysis

a. Ipconfig: this displays the host's network configuration (IP Address, gateway, DNS). It is a routine administrative command, but it is also a common first step when a user or intruder checks which network the machine is on
b. Cd Downloads: the user moved into the downloads Directory, where *secret_file.docx* was presumably located.
c. Copy secret_file.docx F: this is the most significant finding. It shows an attempt to copy a file named *secret_file.docx* to drive F:. drive letters assigned after C: are commonly removable media (USB drive) or mapped/secondary volumes. The sequence of locating the file, navigating to its folder and copying it to another volume is consistent with *data staging or possible exfiltration.*
d. Reset flag and ? entries: the *reset* flag means the history buffer was reinitialized.

5.3 Limitations

- *cmdscan* recordes command text only. It has no timestamps, no output and no proof that the command succeded
- it does not identify which user account ran the commands
- it does not confirm that F: was a USB device or that the file copied successfully.
- The nil(?) entries cannot be relied upon.

5.4 Recommended Corroboration

- Pslist: confirm cmd.exe (PID 3988), its parent process and start time (UTC)
- Consololes: recover te command output, which may show weather the copy succeded (e.g. 1 file(s)copied)
- Filescan: to locate secret file.docx and its timestamps
- Netscan: chechk connections tied to the sshd.exe process.

6. Removable-media evidence (Question 6)

6.1 Device history (SYSTEM hive)

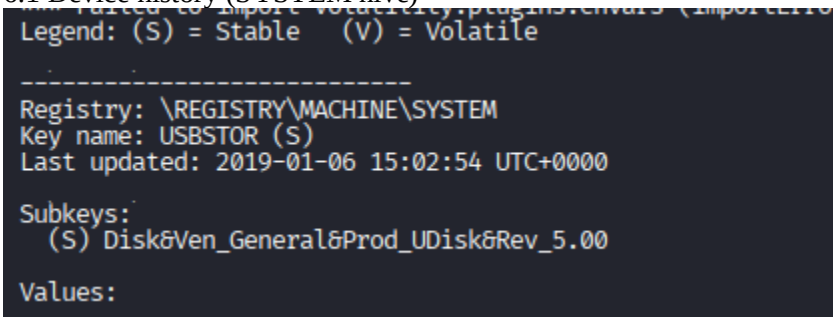


Figure 13 (E13) Device history. The USBSTOR key records every USB mass storage device that has been connected to the system. Its presence confirms that at least one removable storage device, ageneric "General UDisk" flash drive was attached to this machine. This supports the earlier cmdscan finding (Copy secret_file.docx F:), since F: is consistent with a US drive.

6.2 Device History (MountedDevices)

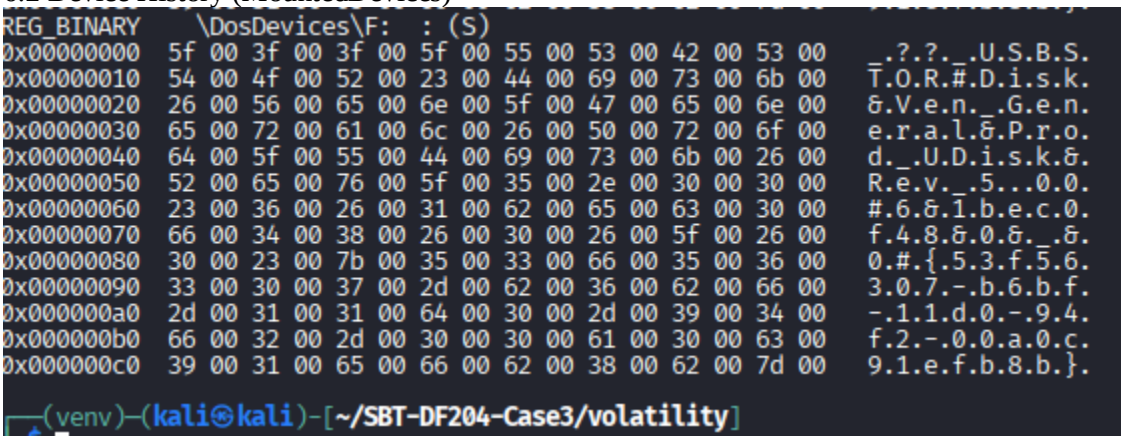


Figure 14 (E14). Mounted Devices evidence. The Drive letter F: was assigned to a USB storage device, the "General UDisk" flash drive. This matches the device found in USBSTOR (section 6.1). it links the three artifacts: the cmdscan entru copy secret_file.docx F: targeted a removable USB drive, not a fixed or network volume

7. ShellBags (Question 7)

```
┌──(venv)─(kali㉿kali)-[~/SBT-DF204-Case3/volatility]
└─$ python2 vol.py -f ~/SBT-DF204-Case3/memdumpWin7.mem --profile=Win7SP1x86_23418 shellbags --output=html --output-file=shellbags.html
Volatility Foundation Volatility Framework 2.6.1
```

8. Findings

Q3. User and account artefacts

Accounts and SIDs found, NTUSER.DAT / UsrClass.dat hive paths and offsets. State that a profile is not proof of a person. (E04)

Q4. Process and network evidence

Relevant PIDs, names, parent/child relationships, start/exit times. Netscan entries: local endpoint, foreign endpoint, state, owner PID. Note any difference between pslist and psscan. No endpoint was contacted. (E05, E06)

Q5. Command history

[Commands from cmdscan / consoles, owning process and PID, offsets, drive or path references. State whether any output confirms success, or that success is not shown.] (E07)

Q6. Removable-media evidence

[USBSTOR device details and last-write time; MountedDevices drive letter; DeviceClasses; MountPoints2 GUID and the owning user hive. For each, state what it establishes and what it does not.] (E08, E09, E10)

Conclusion

The artifacts above agree with one another. The memory images above shows that the user session IEUser copied *Secret_file.docx from C:\Users\IEUser\Downlloads to drive F:, that the* Drive F: was mapped to a generic USB flash drive (Generic UDisk 5.00), combined with the command history. This supports conclusion that *Secret_file.docx* was copied to removable media, Confirming the copy.